BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

HEAT AND MATERIAL BALANCE AND LINE SIZING

DOCUMENT NO. : BFWH-V1-PE-HMB-001

DISCIPLINE : Process

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Streams 1–8 of BFW-PFD-100 for the design case and three off-design cases (maximum load, cold BFW, low-load stall), the worked mass and energy balance, the line list of BFWH-V1-PID-001 rev D with velocity, ρv^2 and Darcy pressure drop, the P-101A/B pump and system curves with NPSH, and the normal / rated / design conditions handed to the datasheets. All numbers from the simulator model and the Design tab; assumptions are labelled.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

CONTENTS

1 Purpose and scope 2

2 Design and off-design cases 3

3 Heat and material balance 3

4 Worked mass and energy balance at design 5

5 Line list and line sizing 6

6 Pump hydraulics and NPSH 7

7 Design cases and margins 7

8 Checks 7

9 Engineering queries 8

Notes 8

1 Purpose and scope

Process engineering turns the process concept into the numbers every other discipline designs to: PFD → heat and material balance (HMB) → line list and line sizing → datasheet inputs and design cases. This document covers E-101 (BFW steam heater), P-101A/B, TK-101 and the steam, condensate and vent lines on BFWH-V1-PID-001 rev D. The sizing of E-101 itself is on the Design tab and is used here, not repeated.

Pressures are kPa g. Liquid volumetric flows are on the model basis of 998 kg/m³. Values marked **assumed**, **TBC** or **HOLD** are not design data.

- Inputs: simulator model design snapshot; Design-tab operating-window runs (DS_OW); PFD BFW-PFD-100; P&ID BFWH-V1-PID-001 rev D.
- Outputs to: BFWH-V1-ME-EQL-001 (equipment list and datasheets), relief and instrument documents, P&ID rev E (line sizes).

2 Design and off-design cases

Design is the model lined out at design. The three off-design cases are model runs: maximum load (BFW raised until FV-104 is 100 % open — the rated case), cold BFW (inlet lowered until the shell needs the spare steam) and low load (BFW lowered until condensate starts to back up into the shell). Recycle and pump discharge pressure come from the model hydraulics at each case's FIC-101 output.

Case	BFW m³/h	Recycle m³/h	T in → out °C	Steam kg/h	Shell kPa g / °C	FV-104 %	FIC-101 OUT %	PI-106 kPa g	PDI-109 kPa	Hotwell %	BFW duty kW	Steam×λ kW	Closure %
Design (normal)	100.0	10.7	18.0 → 70.0	9,935	105.0 / 121.3	50.0	55.0	662.4	60.0	5.0	6,040	6,040	−0.00
Maximum load (rated)	121.2	2.4	18.0 → 69.9	12,161	158.5 / 128.9	100.0	89.2	571.6	88.2	5.0	7,308	7,308	0.00
Cold BFW	100.0	10.7	10.7 → 70.0	11,468	156.3 / 128.6	90.2	55.0	662.4	60.0	5.0	6,888	6,895	0.10
Low load (stall)	72.3	17.7	18.0 → 70.0	7,074	46.1 / 110.8	27.5	31.9	787.2	31.4	12.1	4,367	4,367	0.01

3 Heat and material balance

Enthalpy reference: liquid water at 0 °C, h = $c_p T$ with c_p 4.19 kJ/kg K (model constant); saturated steam h = $c_p T_{sat}$ + λ(T_{sat}). Stream 1 pressure is the pump suction basis K.Ptk = 30 kPa g (TK-101 static head, assumed). Stream 4 pressure = PI-106 − tube ΔP.

Design (normal) — model snapshot at design (snapDesign)

No.	Stream	Phase	m³/h	kg/h	T °C	P kPa g	h kJ/kg	H kW
1	BFW from TK-101 to the pumps	liquid	110.7	110,473	18.0	30.0	75.4	2,314
2	Recycle to TK-101 through FV-101B	liquid	10.7	10,673	18.0	662.4	75.4	224
3	BFW into the E-101 tubes (FIC-101, TI-103)	liquid	100.0	99,800	18.0	662.4	75.4	2,091
4	Heated BFW to the deaerator (TT-101)	liquid	100.0	99,800	70.0	602.4	293.3	8,131
5	Flash steam from the header (PI-105)	sat. steam	—	9,935	133.3	195.0	2,707.1	7,471
6	Steam into the shell after FV-104 (FIC-104)	sat. steam	—	9,935	121.3	105.0	2,696.9	7,443
7	Condensate through the trap to the return header	liquid → flashing	—	9,935	119.3	30.0	500.0	1,380
8	Losses: vent, trap bypass, drain, live steam passing the trap	—	—	0	—	—	—	—

Energy balance: BFW $\dot{m} c_p \Delta T$ = 27.722 × 4.19 × (70.00 − 18.00) = 6,040.1 kW; steam × λ = 2.7598 × 2,188.6 = 6,040.1 kW; closure −0.00 %.

Maximum load (rated) — model run: BFW raised until FV-104 is 100 % open (DS_OW fCV)

No.	Stream	Phase	m³/h	kg/h	T °C	P kPa g	h kJ/kg	H kW
1	BFW from TK-101 to the pumps	liquid	123.6	123,378	18.0	30.0	75.4	2,585
2	Recycle to TK-101 through FV-101B	liquid	2.4	2,370	18.0	571.6	75.4	50
3	BFW into the E-101 tubes (FIC-101, TI-103)	liquid	121.2	121,008	18.0	571.6	75.4	2,535
4	Heated BFW to the deaerator (TT-101)	liquid	121.2	121,008	69.9	483.4	292.8	9,843
5	Flash steam from the header (PI-105)	sat. steam	—	12,161	133.0	192.5	2,706.9	9,144
6	Steam into the shell after FV-104 (FIC-104)	sat. steam	—	12,161	128.9	158.5	2,703.5	9,132
7	Condensate through the trap to the return header	liquid → flashing	—	12,161	126.9	30.0	531.6	1,796
8	Losses: vent, trap bypass, drain, live steam passing the trap	—	—	0	—	—	—	—

Energy balance: BFW $\dot{m}c_p\Delta T$ = 33.613 × 4.19 × (69.89 − 18.00) = 7,308.2 kW; steam × λ = 3.3781 × 2,163.5 = 7,308.4 kW; closure 0.00 %.

Cold BFW — model run: BFW inlet lowered until the shell needs the spare steam (DS_OW tin90)

No.	Stream	Phase	m³/h	kg/h	T °C	P kPa g	h kJ/kg	H kW
1	BFW from TK-101 to the pumps	liquid	110.7	110,473	10.7	30.0	44.8	1,376
2	Recycle to TK-101 through FV-101B	liquid	10.7	10,673	10.7	662.4	44.8	133
3	BFW into the E-101 tubes (FIC-101, TI-103)	liquid	100.0	99,800	10.7	662.4	44.8	1,243
4	Heated BFW to the deaerator (TT-101)	liquid	100.0	99,800	70.0	602.4	293.3	8,131
5	Flash steam from the header (PI-105)	sat. steam	—	11,468	133.1	193.3	2,706.9	8,623
6	Steam into the shell after FV-104 (FIC-104)	sat. steam	—	11,468	128.6	156.3	2,703.2	8,611
7	Condensate through the trap to the return header	liquid → flashing	—	11,468	126.6	30.0	530.4	1,690
8	Losses: vent, trap bypass, drain, live steam passing the trap	—	—	0	—	—	—	—

Energy balance: BFW $\dot{m}c_p\Delta T$ = 27.722 × 4.19 × (70.00 − 10.70) = 6,888.1 kW; steam × λ = 3.1856 × 2,164.5 = 6,895.0 kW; closure 0.10 %.

Low load (stall) — model run: BFW lowered until condensate starts to back up into the shell (DS_OW stall)

No.	Stream	Phase	m³/h	kg/h	T °C	P kPa g	h kJ/kg	H kW
1	BFW from TK-101 to the pumps	liquid	90.0	89,828	18.0	30.0	75.4	1,882
2	Recycle to TK-101 through FV-101B	liquid	17.7	17,673	18.0	787.2	75.4	370
3	BFW into the E-101 tubes (FIC-101, TI-103)	liquid	72.3	72,155	18.0	787.2	75.4	1,512
4	Heated BFW to the deaerator (TT-101)	liquid	72.3	72,155	70.0	755.8	293.3	5,879
5	Flash steam from the header (PI-105)	sat. steam	—	7,074	133.6	197.5	2,707.3	5,320
6	Steam into the shell after FV-104 (FIC-104)	sat. steam	—	7,074	110.8	46.1	2,687.1	5,280
7	Condensate through the trap to the return header	liquid → flashing	—	7,074	108.8	30.0	456.0	896
8	Losses: vent, trap bypass, drain, live steam passing the trap	—	—	0	—	—	—	—

Energy balance: BFW $\dot{m}c_p\Delta T$ = 20.043 × 4.19 × (70.00 − 18.00) = 4,367.0 kW; steam × λ = 1.9650 × 2,222.6 = 4,367.5 kW; closure 0.01 %.

4 Worked mass and energy balance at design

Relation	Numbers	Result
BFW mass flow $\dot{m} = F\rho/3600$	100.0 × 998 / 3600	27.722 kg/s
Duty $Q = \dot{m}c_p(T_{out} - T_{in})$	27.722 × 4.19 × (70.0 − 18.0)	6,040.1 kW
Shell $T_s = T_{sat}(P_s)$	P_s = 105.0 kPa g	121.32 °C
Latent heat $\lambda(T_s)$	model steam table	2,188.6 kJ/kg
Steam $W = Q/\lambda$	6,040.1 / 2,188.6 × 3600	9,935 kg/h
Steam per tonne of BFW	9,935 / 99.8	99.6 kg/t
Mass: stream 1 = 2 + 3	110.69 = 10.69 + 100.00 m³/h	closes
Mass: steam 6 = condensate 7	9,935.4 = 9,935.4 kg/h	closes
Pump $P_d = P_{tk} + H_0 - a(Q + Q_r)^2$	30 + 1000 − 0.03 × 110.69²	662.4 kPa g
Tube $\Delta P = k_t F^2$	0.006 × 100.0²	60.0 kPa
Line to deaerator $k_l F^2$	0.004 × 100.0²	40.0 kPa
FV-101A $\Delta P = P_d - P_b$ − tube − line	662.4 − 250 − 60.0 − 40.0	312.4 kPa
Flash after the trap $x = c_p(T_c - T_{ret})/\lambda(T_{ret})$	4.19 × (119.32 − 107.37) / 2,233.7	2.24 % = 223 kg/h

The balance closes on λ alone: the model drains condensate at T_s − 2 K and counts only latent heat; the subcooling is 23 kW (0.38 %). Throttling through FV-104 is isenthalpic, so real steam leaves the valve slightly superheated (stream 5 vs 6 enthalpy, 10.2 kJ/kg); the model treats it as saturated.

5 Line list and line sizing

Every line number on BFWH-V1-PID-001 rev D at the design and maximum-load cases. ID by schedule 40 (assumed). ΔP per 100 m of straight pipe by Darcy with the Haaland friction factor, roughness 0.045 mm (assumed); steam at the line's own density (incompressible over 100 m, indicative); two-phase homogeneous after the trap; steam viscosity 13 µPa s (assumed). Criteria are typical practice: pump suction 0.5–1.5 m/s, liquid discharge 1–3 m/s, saturated steam 15–40 m/s, condensate to the trap ≤ 1.5 m/s, two-phase after the trap ≤ 15 m/s. Vent at 1 % of steam (assumed continuous-vent rate).

Line	From → to	Phase	ID mm	Design flow	v m/s	ρv² Pa	ΔP kPa /100 m	Max-load flow	v m/s	ρv² Pa	ΔP kPa /100 m	Typical criterion	Checked at m/s	OK
8"-BW-100	TK-101 → P-101A/B suction (V-106, V-107)	liquid	202.7	110.7 m³/h	0.95	905	3.86	123.6 m³/h	1.06	1,129	4.74	pump suction liquid 0.5–1.5 m/s	1.06 (maximum load)	✓
6"-BW-101	P-101A/B discharge (NRV-101/102) → FT-101 → FV-101A → E-101 tubes	liquid	154.1	100.0 m³/h	1.49	2,215	12.47	121.2 m³/h	1.81	3,257	17.96	liquid discharge 1–3 m/s	1.81 (maximum load)	✓
3"-BW-102	pump discharge → FV-101B → TK-101 (recycle)	liquid	77.9	10.7 m³/h	0.62	387	5.69	2.4 m³/h	0.14	19	0.39	liquid discharge 1–3 m/s at the rated recycle (FV-101B wide open)	1.72 (rated: FV-101B wide open)	✓
6"-BW-103	E-101 tube outlet (TT-101, TT-108) → deaerator (boiler house)	liquid	154.1	100.0 m³/h	1.52	2,263	11.74	121.2 m³/h	1.84	3,326	17.07	liquid discharge 1–3 m/s	1.84 (maximum load)	✓
12"-LS-101	flash steam header branch → XV-100 → FE-104 → FV-104	sat. steam	303.2	9,935 kg/h	24.19	925	2.15	12,161 kg/h	29.84	1,396	3.20	saturated steam 15–40 m/s	29.84 (maximum load)	✓
14"-LS-102	FV-104 outlet → E-101 shell steam inlet	sat. steam	333.3	9,935 kg/h	27.90	882	1.86	12,161 kg/h	27.64	1,070	2.22	saturated steam 15–40 m/s	27.90 (design)	✓
1"-VT-101	E-101 shell, far end (air vent) → V-104 → vent to safe location	sat. steam	26.6	99 kg/h	43.69	2,163	97.85	122 kg/h	43.28	2,623	117.27	steam / vent gas ≤ 40 m/s at the continuous-vent rate (1 % of steam, assumed)	43.69 (design)	✗ EQ-10
3"-SC-101	E-101 hotwell → V-101 → Y-101 → ST-101	sat. condensate	77.9	9,935 kg/h	0.61	355	4.35	12,161 kg/h	0.76	535	6.43	saturated condensate to the trap ≤ 1.5 m/s (drains freely, no flashing)	0.76 (maximum load)	✓
4"-SC-102	ST-101 → V-102 → condensate return header 30 kPa g	two-phase	102.3	9,935 kg/h	10.42	3,501	31.56	12,161 kg/h	20.54	8,447	73.70	two-phase condensate after the trap ≤ 15 m/s (homogeneous)	20.54 (maximum load)	✗ EQ-11

Flash after the trap: condensate at 119.3 °C flashes to 107.4 °C at the 30 kPa g return — 2.24 % (223 kg/h) at design and 3.66 % (445 kg/h) at maximum load. The flash steam fills most of the volume of the line downstream of the trap, so it is sized as two-phase.

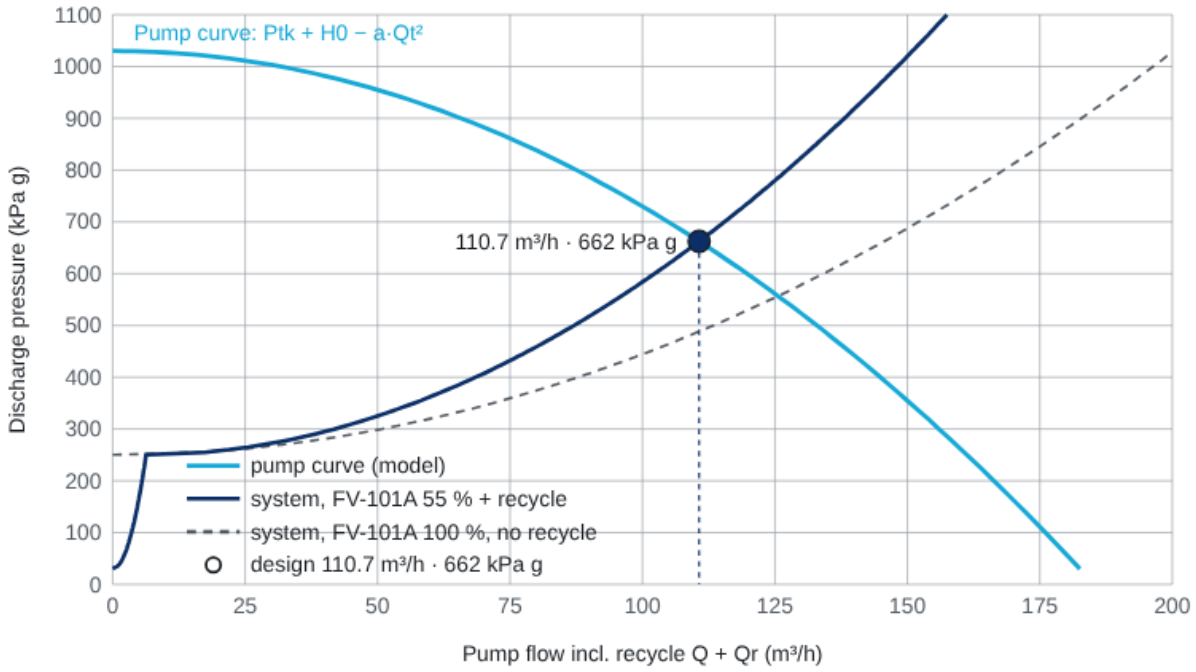
3"-BW-102 is checked at its rated flow, FV-101B wide open (29.5 m³/h); at design it carries 10.7 m³/h.

6 Pump hydraulics and NPSH

Model pump curve $P_d = P_{tk} + H_0 - a(Q + Q_r)^2$, P_{tk} 30 kPa g, H_0 1000 kPa, a 0.03 kPa/(m³/h)²; shut-off 1,030 kPa g (102.1 m). The system is FV-101A + E-101 + line to the deaerator (back-pressure 250 kPa g) in parallel with the FV-101B recycle; FIC-101 output x opens FV-101A and closes FV-101B (100 − x).

Design: FV-101A 55.0 %, 110.7 m³/h at 662.4 kPa g, head 64.6 m, hydraulic 19.4 kW, absorbed 27.8 kW (η 0.7, assumed). Rated (maximum load): FV-101A 89.2 %, 123.6 m³/h at 571.6 kPa g, head 55.3 m, absorbed 26.6 kW. FV-101A wide open: 125.6 m³/h at 557 kPa g.

NPSH available (assumed: TK-101 vented, 101.3 kPa; minimum level 3.06 m above the pump centreline = K.Ptk; suction 25 m equivalent of 8" plus K = 2): NPSHa = 10.34 + 3.06 − 0.21 − 0.191 = **13.01 m** at design and 12.96 m at rated. NPSHr 4 m assumed, HOLD (vendor); acceptance NPSHa ≥ 1.3 × NPSHr.



P-101A/B model curve, system curve at the design FV-101A opening (with recycle) and with FV-101A wide open, design operating point.

7 Design cases and margins

Normal = design case; rated = steam-limited maximum load; mechanical design from the Design tab (shell 350 kPa g = PSV-100 set, tube side 1 100 kPa g, assumed) with design temperature from the relieving temperature plus margin (assumed).

Quantity	Normal	Rated	Rated / normal	Mechanical design
BFW flow, m³/h	100.0	121.2	1.212	—
Pump flow incl. recycle, m³/h	110.7	123.6	1.117	—
Pump discharge, kPa g	662.4	571.6	—	1,100 tube side (assumed); shut-off 1,030
Steam, kg/h	9,935	12,161	1.224	—
Duty, kW	6,040	7,308	1.210	—
Shell pressure, kPa g	105.0	158.5	—	350 (= PSV-100 set, assumed)
Shell temperature, °C	121.3	128.9	—	180 (relieving 151.1 + 25 K, assumed)
Tube ΔP (PDI-109), kPa	60.0	88.2	1.470	—
BFW outlet, °C	70.0	69.9	—	180 (assumed)

8 Checks

3 checks fail on purpose; each is an engineering query (section 9).

Check	Values	Result
Every HMB stream of the design case = model snapshot (FIC-101, FV-101B, TIC-101, FIC-104, PI-100, PI-105, PI-106, Q)	equal	✓

Check	Values	Result
Energy balance closes, Design (normal)	6,040 vs 6,040 kW (−0.00 %)	✓
Energy balance closes, Maximum load (rated)	7,308 vs 7,308 kW (0.00 %)	✓
Energy balance closes, Cold BFW	6,888 vs 6,895 kW (0.10 %)	✓
Energy balance closes, Low load (stall)	4,367 vs 4,367 kW (0.01 %)	✓
Mass balances (1 = 2 + 3; 6 = 7)	close	✓
Design steam = Design tab sizing	9,935 vs 9,935 kg/h	✓
Model hydraulics reproduce the Maximum load (rated) BFW flow	121.24 vs 121.25 m³/h	✓
Model hydraulics reproduce the Cold BFW BFW flow	100.00 vs 100.00 m³/h	✓
Model hydraulics reproduce the Low load (stall) BFW flow	72.26 vs 72.30 m³/h	✓
Pump point on the model curve (design)	110.7 m³/h at 662.4 kPa g = PI-106	✓
Pump shut-off ≤ tube-side design	1,030 ≤ 1,100 kPa g	✓
NPSHa ≥ 1.3 × NPSHr (assumed 4 m), rated	12.96 m ≥ 5.20 m	✓
8"-BW-100 within criterion	1.06 m/s (maximum load)	✓
6"-BW-101 within criterion	1.81 m/s (maximum load)	✓
3"-BW-102 within criterion	1.72 m/s (rated: FV-101B wide open)	✓
6"-BW-103 within criterion	1.84 m/s (maximum load)	✓
12"-LS-101 within criterion	29.84 m/s (maximum load)	✓
14"-LS-102 within criterion	27.90 m/s (design)	✓
1"-VT-101 within criterion	43.69 m/s (design)	✗ EQ-10 raised
3"-SC-101 within criterion	0.76 m/s (maximum load)	✓
4"-SC-102 within criterion	20.54 m/s (maximum load)	✗ EQ-11 raised
6"-BW-101 design velocity vs P&ID note 2	1.49 vs 1.5 m/s (±10 %)	✓
12"-LS-101 design velocity vs P&ID note 2	24.19 vs 24 m/s (±10 %)	✓
14"-LS-102 design velocity vs P&ID note 2	27.90 vs 28 m/s (±10 %)	✓
3"-SC-101 design velocity vs P&ID note 2	0.61 vs 0.6 m/s (±10 %)	✓
FIC-101 L alarm warns before the stall	80 > 72.3 m³/h	✓
Rated steam below the FIC-104 H alarm	12,161 < 13,000 kg/h	✓
Rated tube ΔP below the PDI-109 H alarm	88.2 vs 80 kPa	✗ EQ-12 raised
Cold-BFW steam below the FIC-104 H alarm	11,468 < 13,000 kg/h	✓

9 Engineering queries

EQ	Subject	Query	Status
EQ-10	1"-VT-101 vent line velocity	At the continuous-vent rate of 1 % of the design steam (99 kg/h, Design tab) the 1" vent line runs at 43.7 m/s, above the typical 40 m/s; with V-104 wide open for purging the line is choked. Confirm the vent-rate basis (rev B RO-101 orifice or rev C AAV-101) and consider 1½"-VT-101 (18.5 m/s).	Open
EQ-11	4"-SC-102 two-phase velocity after the trap	Condensate flashes 2.24 % at design and 3.66 % at maximum load (shell 158.5 kPa g) when it drops to the 30 kPa g return. Homogeneous velocity in 4"-SC-102 is 10.4 m/s at design but 20.5 m/s at maximum load, above the typical 15 m/s. Confirm 6"-SC-102 (9.1 m/s at maximum load) or accept the higher velocity with an erosion and hammer review; confirm the return-header back-pressure.	Open
EQ-12	PDI-109 high alarm below the rated-case tube ΔP	With clean tubes the rated (maximum-load) case gives a tube ΔP of 88.2 kPa, above the PDI-109 H alarm of 80 kPa, so the alarm would stand whenever the unit runs at its rated load. Propose a flow-compensated alarm on ΔP/F² or a set point above the rated value; Instrumentation to confirm.	Open
EQ-13	TK-101 elevation and minimum level for NPSH	NPSH available (13.0 m) uses the model's suction basis K.Ptk = 30 kPa g as 3.06 m of minimum liquid level above the pump centreline, with an assumed suction line of 25 m equivalent length. Confirm the TK-101 elevation, minimum operating level and suction layout; NPSHr is HOLD for the pump vendor.	Open

Notes

- Numbers are generated from the BFW Steam Heater Ver 1 simulator (data.json of the Process engineering tab); the tab and this document share them.
- Off-design cases are the Design tab operating-window runs; line sizes on the P&ID rev D are TBC until the queries are closed.
- Do not use for construction.